Natural Units, the Cosmological Hierarchy, and a Ramanujan–Heegner Numerical Curiosity

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(Dated: July 15, 2026)

Buckingham's Π-theorem provides a linear-algebraic account of natural units. The constants c, $\hbar$, and G form a dimensionally independent basis for the mechanical dimensions M, L, and T. Introducing the Hubble parameter H produces one dimensionless invariant, $\hbar G H^2/c^5$. Depending on the chosen natural-unit basis, the observed hierarchy of order 10^{-122} may appear in H^2, G, or $\Lambda\ell_{\mathrm{P}}^2$. A numerical curiosity then follows: the inverse seventh power of the Ramanujan constant $\mathcal{R} = e^{\pi\sqrt{163}}$ is also approximately 10^{-122}. The numbers 7 and 163 are both Heegner numbers, though no physical mechanism is implied.

I. BUCKINGHAM'S THEOREM

Buckingham's Π-theorem states that a physical relation involving n dimensional quantities constructed from r independent base dimensions may be expressed using $n-r$ independent dimensionless combinations.

For relativistic gravitation, take the mechanical base dimensions to be $[M]$, $[L]$, $[T]$. Represent each physical quantity by its exponent vector in the ordered basis (M, L, T). The dimensions of the speed of light, the reduced Planck constant, and Newton's gravitational constant are

$$\begin{aligned} [c] &= LT^{-1}, \\ [\hbar] &= ML^2T^{-1}, \\ [G] &= M^{-1}L^3T^{-2}. \end{aligned}$$

Their dimensional exponent vectors are therefore

$$\boldsymbol{d}_c = \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}, \qquad \boldsymbol{d}_\hbar = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}, \qquad \boldsymbol{d}_G = \begin{pmatrix} -1 \\ 3 \\ -2 \end{pmatrix}.$$

Using these vectors as columns gives the dimensional matrix

$$D = \begin{pmatrix} 0 & 1 & -1 \\ 1 & 2 & 3 \\ -1 & -1 & -2 \end{pmatrix}.$$

Its determinant is $\det D = -2 \neq 0$. Hence $\mathrm{rank}(D) = 3$, and the three constants c, $\hbar$, and G are dimensionally independent. They form a basis for the three-dimensional mechanical dimensional space. This establishes the linear-algebraic legitimacy of the natural-unit choice $c = \hbar = G = 1$. The constants have not disappeared physically. Their dimensions have been absorbed into the definitions of the units themselves.

II. ADDING THE HUBBLE PARAMETER

The Hubble parameter has dimensions $[H] = T^{-1}$, and hence dimensional vector $\boldsymbol{d}_H = \begin{pmatrix} 0 & 0 & -1 \end{pmatrix}^{\mathsf{T}}$. Adjoining this fourth quantity gives

$$D_{\mathrm{ext}} = \begin{pmatrix} 0 & 1 & -1 & 0 \\ 1 & 2 & 3 & 0 \\ -1 & -1 & -2 & -1 \end{pmatrix}.$$

The rank remains three: $\mathrm{rank}(D_{\mathrm{ext}}) = 3$. Since there are four quantities and three independent base dimensions, $\mathrm{nullity}(D_{\mathrm{ext}}) = 1$. Buckingham's theorem therefore guarantees one independent dimensionless combination. Let $\Pi = c^a \hbar^b G^d H^e$, so dimensional homogeneity amounts to

$$D_{\mathrm{ext}} \begin{pmatrix} a \\ b \\ d \\ e \end{pmatrix} = \mathbf{0} \qquad \Longrightarrow \qquad \begin{pmatrix} a \\ b \\ d \\ e \end{pmatrix} = \begin{pmatrix} -5 \\ 1 \\ 1 \\ 2 \end{pmatrix}.$$

The resulting invariant is

$$\boxed{\Pi = \frac{\hbar G H^2}{c^5}}.$$

Using the Planck time,

$$t_{\mathrm{P}} = \sqrt{\frac{\hbar G}{c^5}},$$

this may be written $\Pi = (Ht_{\mathrm{P}})^2$. For the present Hubble parameter H_0, $H_0 t_{\mathrm{P}} \sim 10^{-61}$, and therefore

$$\boxed{\Pi_0 = \frac{\hbar G H_0^2}{c^5} \sim 10^{-122}}. \tag{1}$$

This small dimensionless number cannot be eliminated through a change of units. A change of basis can only determine which dimensional constant carries it. In Planck units one chooses $c = \hbar = G = 1$, and the invariant then reduces to $\Pi_0 = H_0^2$, so that $H_0^2 \sim 10^{-122}$. One may instead adopt a Hubble-adapted basis, $c = \hbar = H_0 = 1$, so invariant becomes $\Pi_0 = G$, and therefore

$$\boxed{G_{\{c=\hbar=H_0=1\}} \sim 10^{-122}}.$$

Same hierarchy expressed in another dimensional basis.

III. COSMOLOGICAL CONSTANT

Consider de Sitter spacetime and Planck length,

$$H^2 = \frac{\Lambda c^2}{3}. \quad \ell_{\mathrm{P}} = \sqrt{\frac{\hbar G}{c^3}}.$$

Multiplying the de Sitter relation by $\hbar G/c^5$ gives

$$\frac{\hbar G H^2}{c^5} = \frac{\Lambda \hbar G}{3c^3}.$$

Since $\ell_P^2 = \frac{\hbar G}{c^3}$, it follows that

$$\boxed{\Lambda \ell_P^2 = 3\frac{\hbar G H^2}{c^5}}.$$

Thus the same hierarchy may be expressed as

$$\Lambda \ell_P^2 \sim 10^{-122}.$$

IV. A RAMANUJAN–HEEGNER COINCIDENCE

Define the Ramanujan constant $\mathcal{R} = e^{\pi\sqrt{163}}$. Numerically, $\mathcal{R} \approx 2.6253741264 \times 10^{17}$. Its seventh inverse power is

$$\mathcal{R}^{-7} = e^{-7\pi\sqrt{163}} \approx 1.1632 \times 10^{-122}$$

In Hubble-adapted natural units, we have numerical comparison

$$\boxed{G_{\{c=\hbar=H_0=1\}} \sim 10^{-122} \approx \mathcal{R}^{-7}}.$$

The pair $(7, 163)$ is arithmetically conspicuous because both entries are Heegner numbers. The complete set of Heegner numbers is

$$1,\ 2,\ 3,\ 7,\ 11,\ 19,\ 43,\ 67,\ 163.$$

The appearance of 163 in $\mathcal{R} = e^{\pi\sqrt{163}}$ is tied to the class-number-one property of the imaginary quadratic field

$$\mathbb{Q}(\sqrt{-163})$$

and to the theory of complex multiplication of elliptic curves. The exponent 7, though itself a Heegner number, has no established cosmological derivation in this setting.

Consider now the sixth (even) Riemann zeta value

$$\zeta(6) = \sum_{n=1}^{\infty} \frac{1}{n^6} = \frac{\pi^6}{945} \approx 1.017343062.$$

Define the adjusted quantity $\mathcal{R}' = \frac{\mathcal{R}}{\zeta(6)}$, so we have,

$$\frac{1}{(\mathcal{R}')^7} = \left(\frac{\zeta(6)}{\mathcal{R}}\right)^7 = \zeta(6)^7 e^{-7\pi\sqrt{163}} \approx 1.31200 \times 10^{-122}.$$

Euler's Basel evaluation,

$$\zeta(2) = \frac{\pi^2}{6}$$

allows us to write $\pi = \sqrt{6\zeta(2)}$, so $\pi\sqrt{163} = \sqrt{978\,\zeta(2)}$ and the Ramanujan constant may be written without an explicit π as $\mathcal{R} = \exp\left(\sqrt{978\,\zeta(2)}\right)$. The adjusted seventh inverse power becomes

$$\boxed{\left(\frac{\exp\left(\sqrt{978\,\zeta(2)}\right)}{\zeta(6)}\right)^{-7} \approx 1.31200 \times 10^{-122}.}$$

TABLE I. Zeta-adjusted Ramanujan values, $\mathcal{R}'_s = \mathcal{R}/\zeta(s)$.

s	$\zeta(s)$	$(\mathcal{R}'_s)^{-7}$
4	1.082323234	2.02377×10^{-122}
5	1.036927755	1.49934×10^{-122}
6	1.017343062	1.31200×10^{-122}

Of course we can play this game all we like, but the numerology a little lost as the adjusted $\mathcal{R}$ loses its mythical integer status. Alternatively in the name of "inverse cubic attenuation", just multiply R^{-7} by Apéry's constant, $\zeta(3) = \sum_{n=1}^{\infty} \frac{1}{n^3} \approx 1.202056903$.

$$\frac{\zeta(3)}{R^7} = \zeta(3)e^{-7\pi\sqrt{163}} \approx 1.39826 \times 10^{-122}.$$

Appendix A: Error Analysis

The quoted 10^{-122} is an order-of-magnitude statement. Since $G_H = \hbar G H_0^2/c^5 \propto H_0^2$, the fractional uncertainty is

$$\boxed{\frac{\Delta G_H}{G_H} = 2\frac{\Delta H_0}{H_0}}.$$

Using 1 Mpc $= 3.08567758 \times 10^{19}$ km,

$$H_0(\mathrm{s}^{-1}) = \frac{H_0(\mathrm{km\,s^{-1}\,Mpc^{-1}})}{3.08567758 \times 10^{19}}.$$

Thus $H_0^{\mathrm{CMB}} = 67.4 \pm 0.5$ gives $H_0^{\mathrm{CMB}} \approx 2.1843 \times 10^{-18}\ \mathrm{s}^{-1}$, while $H_0^{\mathrm{local}} = 73.04 \pm 1.04$ gives $H_0^{\mathrm{local}} \approx 2.3679 \times 10^{-18}\ \mathrm{s}^{-1}$. Using $\hbar = 1.054571817 \times 10^{-34}$ J s, $G = 6.67430 \times 10^{-11}\ \mathrm{m^3\,kg^{-1}\,s^{-2}}$, $c = 2.99792458 \times 10^8\ \mathrm{m\,s^{-1}}$, Eq. (1) gives

$$\boxed{G_H^{\mathrm{CMB}} = (1.3868 \pm 0.0206) \times 10^{-122}}$$

$$\boxed{G_H^{\mathrm{local}} = (1.6285 \pm 0.0464) \times 10^{-122}}.$$

The relative uncertainties are 1.48% and 2.85%, respectively, so the spread between the two central values is larger than the uncertainty within either measurement. Indeed, mhe midpoint between the two central estimates is

$$\frac{G_H^{\mathrm{CMB}} + G_H^{\mathrm{local}}}{2} = 1.50765 \times 10^{-122},$$

which lies strikingly close to the $\zeta(5)$-adjusted Ramanujan value, being only 0.55% below the midpoint. Something to aim for in the absence of discernible empiricism!